The production of both core data and proprietary data feeds involves relatively high fixed costs and low variable costs.[1778] Fixed costs are composed of, among others, costs to set up infrastructure, regulatory approval costs, software development costs, administrative costs and overhead costs, while variable costs include costs to contract with and establish connectivity to each customer. Importantly, fixed costs of the production of both core data and proprietary data feeds are not specific to the production of data but also support the exchanges' other services such as intermediating trade. In such markets, the firms have additional incentives to increase the number of their customers in order to spread the fixed cost across a larger base of consumers.

(b) Current Structure of Market for Proprietary Market Data Products

In addition to SIP data, the exchanges voluntarily disseminate proprietary data and charge fees for this data.[1779] Proprietary data fees have increased over the past decade.[1780] SIFMA estimates that, for some broker-dealers, data fees charged by some exchanges went up by three orders of magnitude or more between 2010 and 2018.[1781] One commenter disagreed with this estimate by comparing it to a separate estimate obtained for increases in market data revenue of 78.4% for Nasdaq over the same period (including both revenue from exchange data and non-exchange data products).[1782] The Commission does not believe there is necessarily any

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in this section and believes that these insights are sufficient to support our analysis in this section.

1778 See, e.g., Paul M. Romer, *Endogenous Technological Change*, 98 J. POL. ECON. S71–102 (1990) (pointing out that information is fundamentally distinct from other goods because it has a fixed cost of discovery and a near zero cost of replication).

1779 See supra Section V.B.2(a) for details on these proprietary feeds.

1780 Some commenters agreed that fees have increased. See, e.g., Virtu Letter at 2; Clearpool Letter at 2.

1781 See SIFMA Letter; see also Virtu Letter I at 4 (discussing double "dipping" on fees by the exchanges).

1782 See Nasdaq Letter IV at 29.

contradiction from the contrast in these estimates, since fees for some broker-dealers for market data are not the same thing as Nasdaq revenue for market data products, because the latter of these contains revenue from all broker-dealers as well as market participants who are not broker-dealers who purchase data from Nasdaq, and it is possible that not all these entities purchase the same set of data products. Correspondingly, exchanges' revenues from selling proprietary data and connectivity services also increased over the last several years. For example, Budish, et al. (2019) observe that exchanges earn significant revenues from selling proprietary data (a range of $555.4-$623.0M in 2015 by their estimate), as well as connectivity services (a range of $436.8-$484.8M in 2015 by their estimate).[1783] According to NYSE's Form 1 filings, its revenues from data services (including connectivity revenues but excluding SIP data revenues) increased approximately 93% from 2014 to 2018. Similarly, Nasdaq's Form 1 filings show an approximately 21% increase in their revenues from data services (excluding revenues from connectivity services and SIP data revenues).[1784] On the other hand, during the same period, revenues distributed back to NYSE by the exclusive SIPs increased approximately 18% and the revenues distributed back to Nasdaq increased approximately 12%. The exchanges' differences in their reporting of these numbers make it difficult to compare revenue numbers across

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1783 See Eric Budish, et al., *Will the Market Fix the Market? A Theory of Stock Exchange Competition and Innovation*, (Univ. of Chi., Becker Friedman Inst. for Econ., Working Paper No. 2019–72, May 2019), available at https://ssrn.com/abstract=3391461 (Retrieved from SSRN Elsevier database).

1784 Nasdaq submitted estimates for Nasdaq's increase in revenue from exchange and non-exchange data products of 78.4% over the period from 2010 and 2018, and an increase of 62.4% for revenue from connectivity services from 2010 to 2018. See Nasdaq Letter IV at 29.

exchanges. However, for both of these exchanges, their revenues from the proprietary data and connectivity business have been growing faster than the revenues they collect from SIP data.[1785]

Indicia that exchanges may not be subject to robust competition include that many broker-dealers state that even in the face of increasing proprietary data fees they feel compelled to buy proprietary data to be able to provide competitive trading strategies for their clients.[1786]

[1785] According to its 2014 Form 1 filing, NYSE collected approximately $138 million as market data revenues, covered under the "data services fees" income statement line item. According to the notes to NYSE's financial statements, these market data revenues include proprietary data revenues, SIP data revenues, and revenues from connectivity services. NYSE's same revenue line item increased to approximately $236 million by the end of 2018. Whereas during this same time period, the revenues NYSE collected from the exclusive SIPs went from approximately $40 million to approximately $47 million. Nasdaq's 2014 Form 1 filing discloses approximately $206 million in "information services" line item in its income statement. According to the footnotes to its financial statements, this line item includes Nasdaq's market data revenues and redistributed SIP revenues but does not include connectivity service revenues. In its 2018 Form 1 filing, Nasdaq disclosed $242 million in revenues under the same information services line item. During the same time period, Nasdaq's SIP data revenues went up from approximately $76 million to $85 million, a smaller revenue increase relative to its market data revenues.

[1786] See, e.g., SIFMA Letter at 2 ("[W]e do not believe that the SIPs currently provide the necessary data to market participants at the requisite speed to efficiently trade in today's high speed and automated marketplace. As a result, many broker-dealers, asset managers and other market participants are forced to purchase proprietary data feeds from individual exchanges to create a consolidated and robust view of the market, while additionally bearing the economic burden of having to purchase consolidated data from the SIPs. This results in an enormous cost burden on the marketplace and creates a two-tiered market for market data by limiting access to critical market data at the fastest speeds to those who can afford to pay the exorbitant fees charged for it by the exchanges."); MFA Letter at 2 ("Today, the current exclusive SIP model and content of core data does not serve the needs of investors, many of whom must subscribe to the exchanges' proprietary market data feeds at considerable additional cost to trade effectively, while others are forced to rely on inferior information and outdated technology."); State Street Letter at 2 (". . . regulatory obligations and customer expectations related to best execution, transaction cost analysis, transparency and market competition generated further need for data that is unavailable on the SIPs. As a result, market participants have become increasingly dependent on proprietary data feeds

Additionally, some academic research suggests that each particular exchange's proprietary data has no substitutes for some uses of the data and no perfect substitutes for any uses.[1787] For example, Budish et al. (2019) conclude that each exchange has market power with respect to the data products (and the speed technology) specific to that particular exchange because of a lack of substitutes for many applications of their data.[1788]

A commenter stated that there is competition between exchanges for proprietary data products as part of their overall competition for order flow.[1789] While it is true that exchanges compete for order flow, they are the sole source of data from their own exchange. Many market participants use a full view of the market in order to route orders effectively, regardless of whether or not they end up sending the order to any particular exchange. The Commission

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marketed by the exchanges outside of the SIPs."); Capital Group Letter at 2 ("Over the last 15 years, the discrepancy in data elements and latency between proprietary feeds and the consolidated tape has expanded such that the SIP is no longer a realistic tool for institutional investors or broker-dealers in meeting their respective best execution obligations when routing orders."). See also Roundtable Day One Transcript at 198–199 (Joseph Wald, Clearpool) ("Clearpool and other broker-dealers are compelled to purchase exchanges' proprietary data feeds, both to provide competitive execution services to our clients and to meet our best execution obligations due to the content of the information contained in the proprietary data feeds as well as the latency differences between them, which are major and important considerations for brokers.").

1787 These points are supported by some commenters. See, e.g., BestEx Research Letter at 2, 4 ("…exchanges do not compete on market data fees since each is an exclusive provider of their own, indispensable content.") and DOJ Letter at 4, stating that characteristics of proprietary data feeds "…would tend to indicate that Prop Data products lack substitutes, which would in turn enable the exchanges to exercise market power in determining their pricing of these products because they are the only data provider in their own markets."

1788 See Eric Budish et al., supra note 1783. See also Glosten, *Economics of the Stock Exchange Business: Proprietary Market Data*, (Jan. 2020), available at: https://ssrn.com/abstract=3533525 (Retrieved from SSRN Elsevier database).

1789 See, e.g., Nasdaq Letter IV at 48.

understands that some market participants may combine the exclusive SIPs with proprietary feeds in order form a complete view of the market, but this comes with disadvantages, as discussed elsewhere in this release.[1790]

This commenter also stated in reference to the question of competition in the provision of proprietary data that "the Commission's analysis is incomplete and flawed because it fails to appropriately analyze competition between trading platforms, and never considers the all-in price of trading in its discussion."[1791] The Commission does not believe that its analysis presented on this issue is incomplete or flawed for not including the all-in cost of trading because market data and trading services, although related, are not the same thing.[1792] For example, it is conceivable that market participants may want data from an exchange even if they never route orders to that exchange. In such a case, the cost of trading on that exchange is not even relevant to that market participant. Therefore, whether exchanges face robust competition in the market for their proprietary data products can be determined by considering the indicia discussed above (among other things) and without consideration of other costs of trading, which include not only other SRO products, but often products and services provided by additional third-party vendors. Because of these considerations, the Commission also does not believe that the metric offered by one commenter[1793] produced by dividing one exchange group's total revenue by its total dollar trading volume is necessarily relevant to the question of robust competition and pricing in the market for proprietary data products. Specifically, this revenue includes revenue across all

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1790 See supra Section V.B.2(b).

1791 Nasdaq Letter IV at 29.

1792 The relevant analysis was presented in the Proposing Release, 85 FR at Section VI.B.3(b). See Proposing Release, 85 FR at Section VI.B.3(d) for a discussion of trading services and the market for their provision.

1793 See Nasdaq Letter IV at 30.

businesses, not just market data, and the value of considering this revenue on a per trade basis at this exchange group is unclear.

(c) Current Structure of Market for Connectivity Services

Exchanges are exclusive providers of their own connectivity services, and for many market participants, effective trading strategies require connection to many if not all of the exchanges, making their demand for these connectivity services less elastic (i.e., less sensitive to price changes). The Commission examined data on exchange orders that shows that large broker-dealers (as measured, for example, by the number of messages sent to exchanges) connect to all or almost all exchanges.[1794] This is consistent with Roundtable participants' stated view that in order to avoid a competitive disadvantage, market participants have little choice but to purchase direct connectivity services from multiple SROs.[1795]

As mentioned above, the exchanges offer different connectivity options to transmit market data to market participants. These options may include fiber optics connections, wireless microwave connections, and laser transmission, all of which vary in speeds and reliability.[1796] The fastest and more reliable connections (e.g., laser transmission) offer market participants an advantage over other market participants with slower or less reliable connections. Therefore, the

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1794 Based on the sample of audit trail data made available to the Commission by FINRA, firms that are connected to all exchanges account for 76.6% of the message volume (there are 37 such firms out of a total of 327 firms in the sample). Firms that are connected to at least all but one of the exchanges account for 91.6% of the message volume (there are 50 such firms). The FINRA data sample covers the week of December 5, 2016, and includes messages sent to 11 exchanges (NYSE National and Chicago Stock Exchange are not part of this sample).

1795 See Proposing Release, 85 FR at Sections III.C.2.(a), II.A.

1796 See Proposing Release, 85 FR at Section II.A.

Commission believes that the exchanges have incentives to offer multiple levels of connectivity and exchanges can charge higher prices for the fastest connections.

(d) Current Structure of the Market for Trading Services in NMS Stocks

The market for trading services is served by exchanges, ATSs, and liquidity providers. The market relies on competition to supply investors with execution services at efficient prices. These trading venues, which compete to match traders with counterparties, provide a framework for price negotiation and disseminate trading information. The market for trading services in NMS stocks currently consists of 16 national securities exchanges, as well as off-exchange trading venues including wholesalers[1797] and 34 NMS stock alternative trading systems.[1798]

Since the adoption of Regulation NMS in 2005, the market for trading services has become more fragmented. The number of exchanges increased from eight in 2005 to 16 exchanges operating today.[1799] Additionally, the market shares of individual exchanges became less concentrated, with a shift in market shares from some of the bigger and older exchanges to the newer ones.[1800] For instance, from 2005 to 2013, there was a decline in the market share of

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1797 Wholesalers are broker-dealers that pay retail brokers for sending their clients' orders to the wholesaler to be filled internally (as opposed to sending the trade orders to an exchange). Typically, a wholesaler promises to provide price improvement relative to the NBBO for filled orders.

1798 As of November 23, 2020, 34 NMS stock ATSs are operating pursuant to an initial Form ATS-N. A list of NMS stock ATSs, including access to initial Form ATS-N filings that are effective, can be found at https://www.sec.gov/divisions/marketreg/form-ats-n-filings.htm.

1799 See Proposing Release, 85 FR at n.660.

1800 See Letter from Edward T. Tilly, Chairman and Chief Executive Officer, Cboe, to Brent J. Fields, Secretary, Commission (May 25, 2018), at n.9.

trading volume for exchange-listed stocks on NYSE.[1801] At the same time, there was an increase in the market share of newer national securities exchanges such as NYSE Arca, Cboe BYX, and Cboe BZX.[1802]

Additionally, the proportion of NMS stocks trading off-exchange (which includes both internalization and ATS trading) increased; for example, as of July 2020, ATSs alone comprised approximately 10 percent of consolidated dollar volume, and other off-exchange volume totaled approximately 23 percent of consolidated dollar volume.[1803] Aside from trading venues, exchange market makers provide trading services in the securities market. These firms stand ready to buy and sell a security "on a regular and continuous basis at a publicly quoted price."[1804] Exchange market makers quote both buy and sell prices in a security held in inventory, for their own account, for the business purpose of generating a profit from trading with a spread between the sell and buy prices. Off-exchange market makers also stand ready to buy and sell out of their own inventory, but they do not quote buy and sell prices.[1805]

1801 See Securities Exchange Act Release No. 76474 (Nov. 18, 2015), 80 FR 80998, 81112 (Dec. 28, 2015) (Regulation of NMS Stock Alternative Trading Systems Proposing Release).

1802 Id.

1803 Data sources: TAQ and FINRA's OTC Transparency Data weekly summaries (https://otctransparency.finra.org/otctransparency/AtsDownload). Due to FINRA's weekly aggregation, the actual sample is 03/30/2020 through 06/26/2020 (i.e., the last two working days of March are included, and the last two working days of June are excluded).

1804 See Commission, Fast Answers: Market Maker (modified Mar. 17, 2000), available at http://www.sec.gov/answers/mktmaker.html.

1805 See Laura Tuttle, *OTC Trading: Description of Non-ATS OTC Trading in National Market System Stocks*, SECURITIES AND EXCHANGE COMMISSION (Mar. 2014),

Trading venues can rely on the SIP, proprietary feeds, or a combination of both to determine the NBBO for the purposes of trade execution. One commenter observed that over one third of ATSs exclusively rely on the SIP when determining trade prices and that other ATSs used the SIP as a backup and in place of the direct feeds from some exchanges.[1806]

All of these developments increased the competitiveness of the market for trading services in NMS stocks. However, the Commission recognizes that while the market is more competitive, the actual level of competition that any given trading venue faces may depend on multiple factors including the liquidity of a stock as well as the type of trading venue and market participant engaging in the trade.

(e) Broker-Dealers' Competitive Strategies for Trading Services

While many market participants use market data to make investment decisions, not all market participants are equally competitive in how they trade based on this data. Some broker-dealers and other latency sensitive traders utilize sophisticated routing tools to strategically decide how to fill an order, including when and where to submit the order, how to split a larger order (i.e., into how many pieces, or "child orders"[1807]), how large the child order sizes should be, and what order type(s) should be used, e.g., whether to use a market order, limit order, or some other order type. The strategies employed by broker-dealers and other latency sensitive traders in this regard are designed to secure the best possible execution price(s) for an order. For

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available at http://www.sec.gov/dera/staff-papers/white-papers/otc-trading-white-paper-03-2014.pdf.

1806 See BestEx Research Letter at 3.

1807 "Child order" refers to a smaller order that was a piece of a larger "parent" order.

example, the methodologies utilized in trading orders can impact the price of the stock being purchased or sold in a manner that can increase or decrease its execution cost. Commenters stated that the trend towards higher proprietary data fees has had a negative impact on the market, such as making it more difficult for broker-dealers to compete.[1808]

Broker-dealers in particular compete with each other to provide the lowest possible execution costs for their clients (i.e., high execution quality) as quickly as possible.

An example of routing tools as noted above is an SOR. SORs employ the use of algorithms (e.g., by broker-dealers on behalf of a client) designed to optimally send parts of an order (child orders) to various market centers (e.g., exchange and ATSs) so as to optimally access market liquidity while minimizing execution costs. SORs help to determine how to quickly access ("take") available market liquidity before other market participants, and help to determine how to strategically place limit orders to optimize queue priority across various limit order books among exchanges. The ability to optimize queue priority facilitates the ability for a broker to "capture the quoted" spread, i.e., buy on the bid or sell on the offer, while also potentially benefitting from exchange rebates paid to liquidity providers.

The Commission understands that data beyond the NBBO with minimal latency are important inputs[1809] to strategies designed to optimize the ability to access market liquidity and

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1808 See, e.g., Clearpool Letter at 2 ("…of all the issues relating to the costs of trading, the trend toward higher market data fees has had the most negative impact on the securities markets. It remains increasingly difficult for many broker-dealers to compete in the current market environment due, in part, to issues related to the costs associated with trading.").

1809 In addition to such data, the Commission believes that there are also ongoing significant personnel and technological costs to producing a sophisticated, competitive SOR.